KAIZEN MATHS

Lesson starter

Large Data Set Practice

Summary Statistics

Name _____

SA1

Compare the two groups using the summary table. Refer to centre and spread where relevant.

Group	n	Mean	Median	SD	Q1	Q3
Coastal towns	96	76.4	73	21.6	59	91
Inland towns	144	54.8	51	18.1	42	67

SA2

For winter monthly temperature, Q1 = 4.1 and Q3 = 7.2. Find the IQR and interpret it in context.

Show your working

KAIZEN MATHS

Lesson starter

Large Data Set Practice

Summary Statistics

Name _____

SB1

Which group has more variation? Use the standard deviations to justify your answer.

Group	n	Mean	Median	SD	Q1	Q3
Winter months	80	5.8	5.5	2.3	4.1	7.2
Summer months	80	17.6	17.9	2	16.2	18.8

SB2

Compare the two groups using the summary table. Refer to centre and spread where relevant.

Group	n	Mean	Median	SD	Q1	Q3
Winter months	80	5.8	5.5	2.3	4.1	7.2
Summer months	80	17.6	17.9	2	16.2	18.8

Show your working

KAIZEN MATHS

Exit ticket

Large Data Set Practice

Summary Statistics

Name _____

EA1

For inland monthly rainfall, Q1 = 42 and Q3 = 67. Find the IQR and interpret it in context.

Show your working

KAIZEN MATHS

Exit ticket

Large Data Set Practice

Summary Statistics

Name _____

EB1

Which group has more variation? Use the standard deviations to justify your answer.

Group	n	Mean	Median	SD	Q1	Q3
Coastal towns	96	76.4	73	21.6	59	91
Inland towns	144	54.8	51	18.1	42	67

Show your working

KAIZEN MATHS

Exit ticket

Large Data Set Practice

Summary Statistics

Name _____

EA1

For inland monthly rainfall, Q1 = 42 and Q3 = 67. Find the IQR and interpret it in context.

STA-986D-L2 · EA

Show your working

KAIZEN MATHS

Exit ticket

Large Data Set Practice

Summary Statistics

Name _____

EB1

Which group has more variation? Use the standard deviations to justify your answer.

Group	n	Mean	Median	SD	Q1	Q3
Coastal towns	96	76.4	73	21.6	59	91
Inland towns	144	54.8	51	18.1	42	67

STA-986D-L2 · EB

Show your working

Teacher answers & working

Large Data Set Practice · Summary Statistics

STA-986D-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

SA1

Compare the two groups using the summary table. Refer to centre and spread where relevant.

Group	n	Mean	Median	SD	Q1	Q3
Coastal towns	96	76.4	73	21.6	59	91
Inland towns	144	54.8	51	18.1	42	67

Answer:

Coastal towns has the higher mean by 21.6

Working:

Compare the means: Coastal towns has mean 76.4 and Inland towns has mean 54.8.

The difference in means is 21.6.

Compare spread using SD or IQR. Coastal towns has SD 21.6; Inland towns has SD 18.1.

Conclusion in context: Coastal towns has the higher typical value for rainfall, but the spread should also be considered.

SA2

For winter monthly temperature, Q1 = 4.1 and Q3 = 7.2. Find the IQR and interpret it in context.

Answer:

$\text{IQR} = 3.1$

Working:

Use $\text{IQR} = \text{Q3} - \text{Q1}$.

$\text{IQR} = 7.2 - 4.1 = 3.1$.

Interpretation: the middle 50% of values for winter monthly temperature are spread across about 3.1 units.

SB1

Which group has more variation? Use the standard deviations to justify your answer.

Group	n	Mean	Median	SD	Q1	Q3
Winter months	80	5.8	5.5	2.3	4.1	7.2
Summer months	80	17.6	17.9	2	16.2	18.8

Answer:

Winter months

Working:

Standard deviation measures spread around the mean.

Winter months has SD 2.3.

Summer months has SD 2.

The larger standard deviation is 2.3, so Winter months has more variation.

Teacher answers & working

Large Data Set Practice · Summary Statistics

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SB2

Compare the two groups using the summary table. Refer to centre and spread where relevant.

Group	n	Mean	Median	SD	Q1	Q3
Winter months	80	5.8	5.5	2.3	4.1	7.2
Summer months	80	17.6	17.9	2	16.2	18.8

Answer:

Summer months has the higher mean by 11.8

Working:

Compare the means: Winter months has mean 5.8 and Summer months has mean 17.6.

The difference in means is 11.8.

Compare spread using SD or IQR. Winter months has SD 2.3; Summer months has SD 2.

Conclusion in context: Summer months has the higher typical value for mean temperature, but the spread should also be considered.

EA1

For inland monthly rainfall, Q1 = 42 and Q3 = 67. Find the IQR and interpret it in context.

Answer:

$IQR = 25.0$

Working:

Use $IQR = Q3 - Q1$.

$IQR = 67 - 42 = 25.0$.

Interpretation: the middle 50% of values for inland monthly rainfall are spread across about 25.0 units.

EB1

Which group has more variation? Use the standard deviations to justify your answer.

Group	n	Mean	Median	SD	Q1	Q3
Coastal towns	96	76.4	73	21.6	59	91
Inland towns	144	54.8	51	18.1	42	67

Answer:

Coastal towns

Working:

Standard deviation measures spread around the mean.

Coastal towns has SD 21.6.

Inland towns has SD 18.1.

The larger standard deviation is 21.6, so Coastal towns has more variation.